Deeper Learning and 21st Century Skills in a Virginia-Based Elementary STEM Program

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Background

Science, Technology, Engineering, and Mathematics (STEM) education provides opportunities for students to engage in real-life problem solving using embedded 21st Century skills and elements of deeper learning.

STEM education supports future teachers in equipping all students with 21st century skills. In STEM classrooms, lessons covering multiple content areas and 21st century skills promote best instructional practice. In preparation for college and careers in a global economy, STEM activities facilitate the development of relevant workplace skills (e.g., creativity).

This qualitative study analyzes STEM lessons that benefit novice teachers' classrooms. Teachers have called for more professional development in STEM instructional practices (Klimaitis & Mullen, 2020) and mentors can help fill this gap (Mullen & Klimaitis, 2019).

New teachers need lessons and tools for working smarter. Streamlining their workload can help them thrive in the teaching profession. STEM lessons covering multiple content areas and 21st century skills promote best instructional practice for addressing challenges. Fostering deeper learning and engaging underrepresented groups in STEM learning are hurdles.

The Deeper Learning Hub (2019) names six competencies of deeper learning: **content mastery, effective communication, collaboration, critical thinking and problem solving, self-directed learning, and an academic mindset** (sense of belonging and motivation) (p. 1).

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Research Questions

1. What percentage of 21st century skills are included in grade 5 STEM projects?
2. What percentage of deeper learning components are included in grade 5 STEM projects?
3. What percentage of core subjects are included in grade 5 STEM projects?

Theoretical Framework

STEM lessons facilitate opportunities for students to develop sought-after workplace competencies in preparation for college and careers in a global economy. These skills— citizenship, collaboration, communication, creativity, and critical thinking—are called the "5Cs" in Virginia (Virginia Department of Education [VDOE], 2020; also, Mullen & Klimaitis, 2019).

In STEM classrooms, students experience deeper learning through skills-based, relevant activities (Brannen-Sarrategui, 2015). Competencies for deeper learning include content mastery, effective problem solving, self-directed learning, and academic mindset (Deeper Learning Hub, 2019).

STEM and deeper learning share similar elements of 21st Century Skills such as collaboration, communication, critical thinking and creativity (Brannen-Sarrategui, 2015).

Tienken (2017) proposes that "one could conjecture that skills and dispositions that are difficult to offshore – creativity, innovation, and entrepreneurship – will retain value in 2035 and beyond" (p. 46).

Methods

Rubric used to assess embedded opportunities for 21st Century Skills and Deeper Learning.

Projects were also examined for inclusion of core subject areas.

- 124 STEM project samples
- 53 Grade 5 Classroom Teachers
- 14 Elementary Schools

Rubric

STEM DESIGN RUBRIC

Title of STEM Design:

SOL Addressed and Essential Knowledge based on Virginia DOE Website:

Design Brief addressed this Content Area(s): Please Circle: Grade 5 Math, Grade 5 ELA, Grade 5 Science

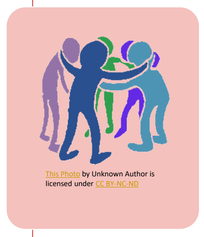
Criteria	Surpasses expectations 3	Meets expectations 2	Does not meet expectations 1	Rating /12
21st Century Skills embedded in STEM Design Critical Thinking Collaboration	Both 21st Century Skills are embedded in STEM Design: Critical Thinking and Collaboration	Only 1 of these two 21st Century Skills are embedded in STEM Design	Neither of these two 21st Century Skills are embedded in STEM Design	
21st Century Skills embedded in STEM Design Creativity Communication	Both 21st Century Skills are embedded in STEM Design: Creativity and Communication	Only 1 of these two 21st Century Skills are embedded in STEM Design	Neither of these two 21st Century Skills are embedded in STEM Design	
Deeper Learning Competencies embedded in STEM Design Content mastery (real world application) Effective Communication Collaboration	All three Deeper Learning Competencies are embedded in STEM Design: Content mastery (real world application); Effective Communication and Collaboration	Only two of these three Deeper Learning Competencies are embedded in STEM Design	Only one of these three Deeper Learning Competencies are embedded in STEM Design	
Deeper Learning Competencies embedded in STEM Design Problem Solving Self-Directed Academic Mindset	All three Deeper Learning Competencies are embedded in STEM Design: Problem Solving; Self-Directed and Academic Mindset	Only two of these three Deeper Learning Competencies are embedded in STEM Design.	Only one of these three Deeper Learning Competencies are embedded in STEM Design	

Results

- Average rubric score: 9.5 (out of 12)
- 21st Century Skills: Critical Thinking & Collaboration = 2.4 (out of 3)
- 21st Century Skills: Creativity & Communication = 2.5 (out of 3)
- Deeper Learning: Content Mastery, Communication & Collaboration = 1.8 (out of 3)
- Deeper Learning: Problem Solving, Self-Directed, Academic Mindset = 2.7 (out of 3)
- At least one core subject was included in all STEM projects; 34% of the STEM projects included two or more core subjects.

*Research conducted with mentor through UCEA.

Findings



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% of Opportunity for 21st Century Skills in STEM Projects

Critical Thinking	90%
Collaboration	50%

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% of Opportunity for 21st Century Skills in STEM Projects

Creativity	90%
Communication	54%

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% of Opportunity for Deeper Learning Components

Content Mastery (Real-World Application)	69%
Communication	54%
Collaboration	50%

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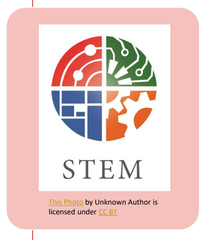
% of Opportunity for Deeper Learning Components

Problem Solving	91%
Self-Directed	93%
Academic Mindset	67%



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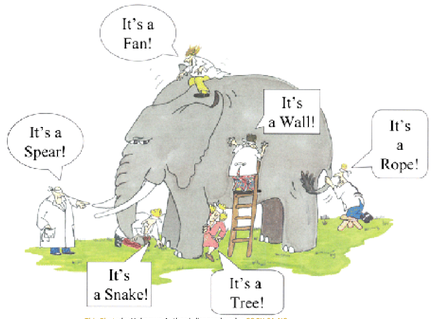
Core Subjects included in STEM Projects

Mathematics	31%
Science	45%
English	42%

Implications

STEM lessons provide opportunities for developing the 5Cs (creativity, etc.) in classrooms, as required by the VDOE (2020). Virginia's graduates cannot be left behind, so they need to be intentionally prepared for the workforce. New teachers and school administrators must gain working curricular knowledge of how best to develop students' capacities in STEM areas.



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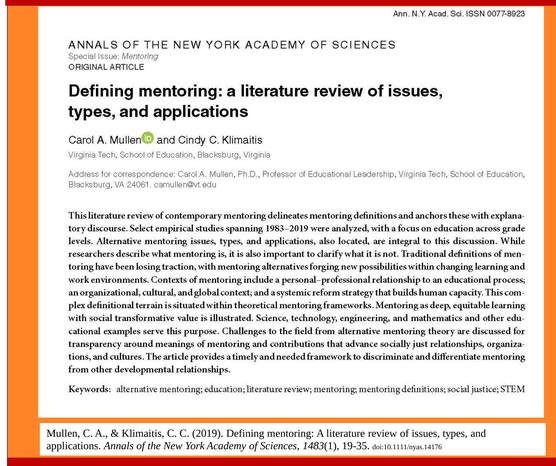
Limitations

Sample size is limited and should not be generalized. Core subject material is based on the Virginia Standards of Learning. STEM projects are from a high-performing suburban school district.

Mentoring



Ann. N.Y. Acad. Sci. ISSN 0077-8923

ANNALS OF THE NEW YORK ACADEMY OF SCIENCES
Special Issue: *Mentoring*
ORIGINAL ARTICLE

Defining mentoring: a literature review of issues, types, and applications

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This literature review of contemporary mentoring delineates mentoring definitions and anchors these with explanatory discourse. Select empirical studies spanning 1983–2019 were analyzed, with a focus on education across grade levels. Alternative mentoring issues, types, and applications, also located, are integral to this discussion. While researchers describe what mentoring is, it is also important to clarify what it is not. Traditional definitions of mentoring have been losing traction, with mentoring alternatives forging new possibilities within changing learning and work environments. Contexts of mentoring include a personal–professional relationship to an educational process; an organizational, cultural, and global context; and a systemic reform strategy that builds human capacity. This complex definitional terrain is situated within theoretical mentoring frameworks. Mentoring as deep, equitable learning with social transformative value is illustrated. Science, technology, engineering, and mathematics and other educational examples serve this purpose. Challenges to the field from alternative mentoring theory are discussed for transparency around meanings of mentoring and contributions that advance socially just relationships, organizations, and cultures. The article provides a timely and needed framework to discriminate and differentiate mentoring from other developmental relationships.

Keywords: alternative mentoring; education; literature review; mentoring; mentoring definitions; social justice; STEM

Mullen, C. A., & Klimaitis, C. C. (2019). Defining mentoring: A literature review of issues, types, and applications. *Annals of the New York Academy of Sciences, 1483*(1), 19-35. doi:10.1111/nyas.14176

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